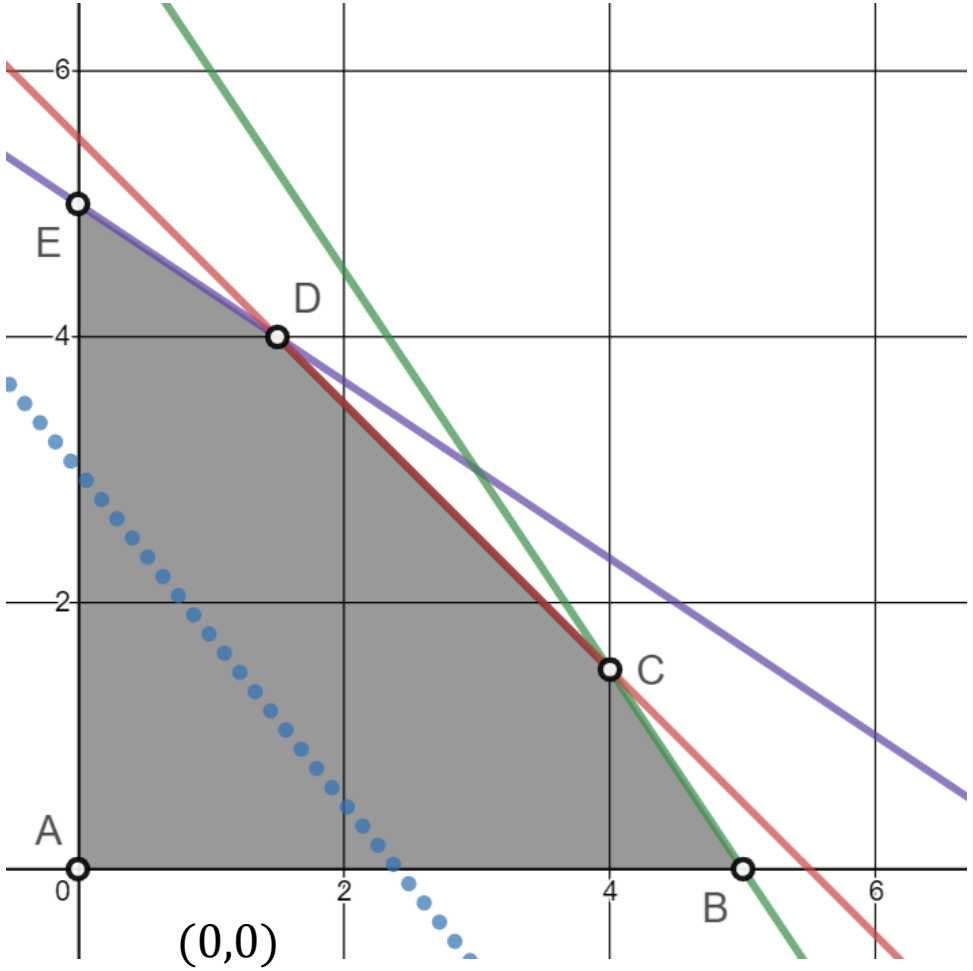
Simplex Method Graphical Demo



maximise

$$P = 5x + 4y$$

subject to:

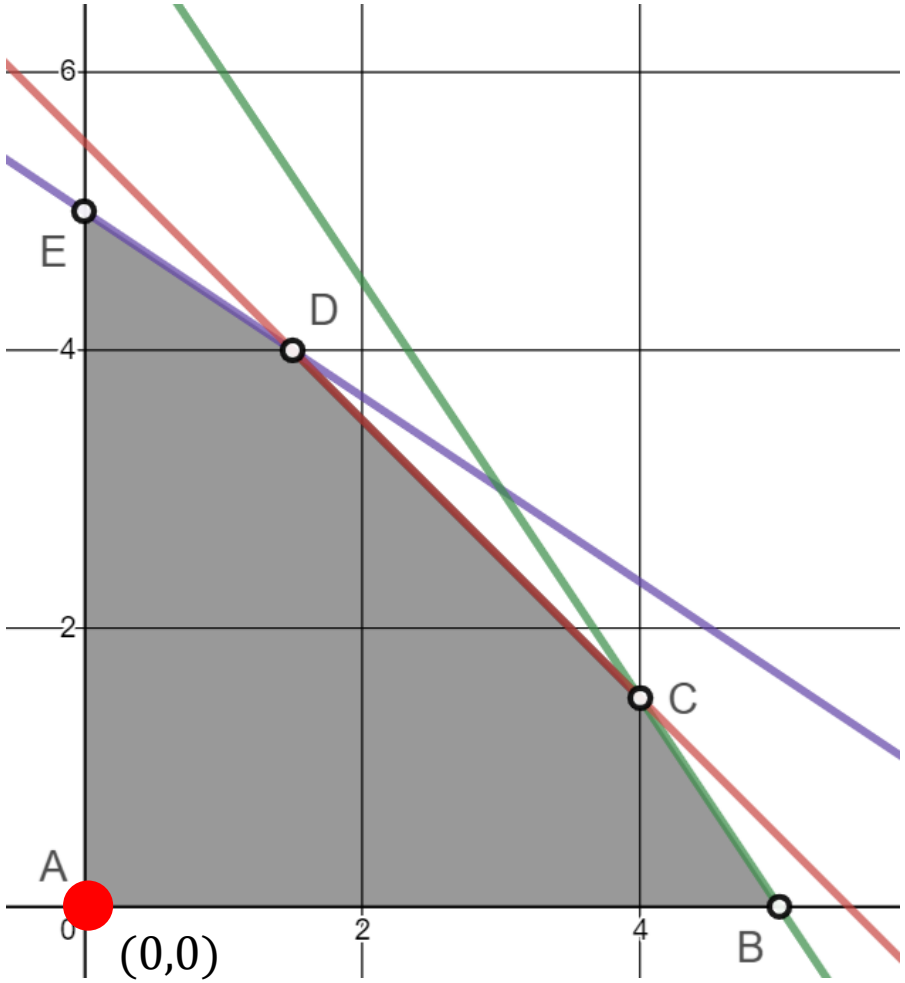
$$2x + 3y \leq 15$$
$$3x + 2y \leq 15$$
$$2x + 2y \leq 11$$

and

$$x, y \geq 0$$

Point A has an objective value of

$$P = 5 \cdot 0 + 4 \cdot 0 = 0$$

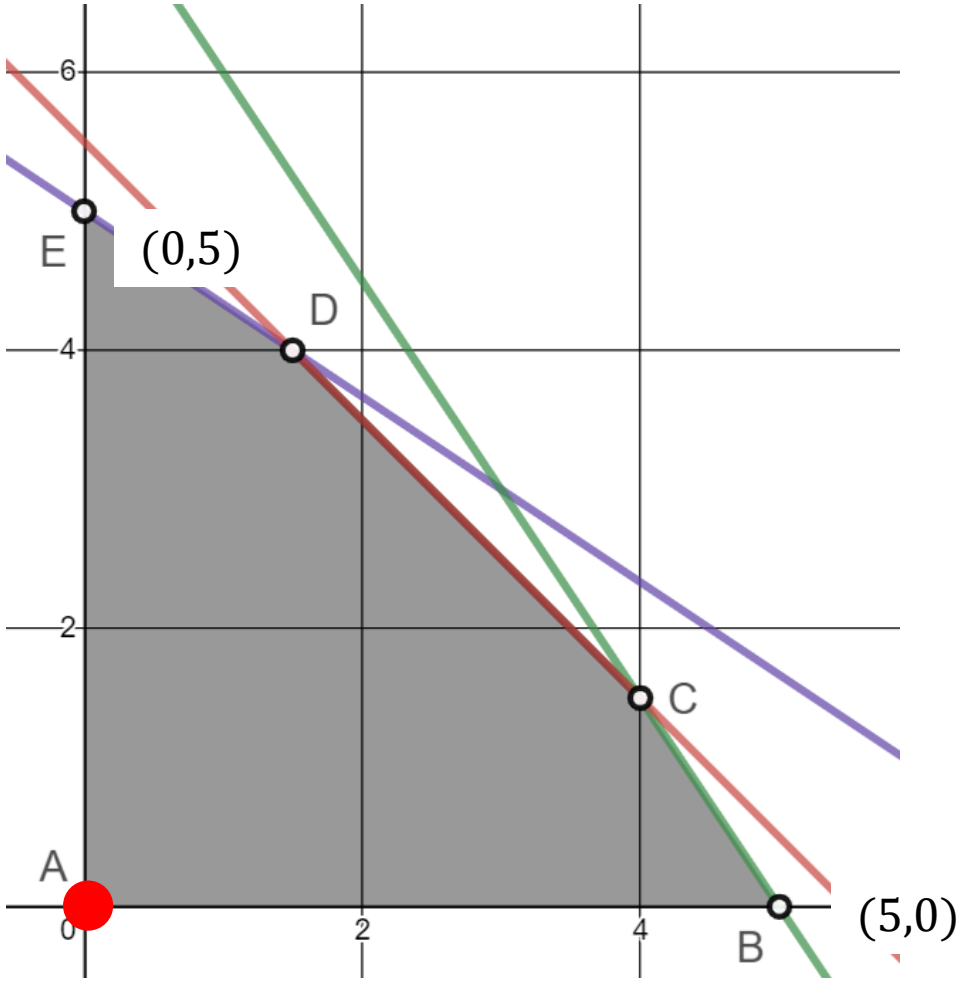


$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$		

Point A has an objective value of

$$P = 5 \cdot 0 + 4 \cdot 0 = 0$$



$$P = 5x + 4y$$

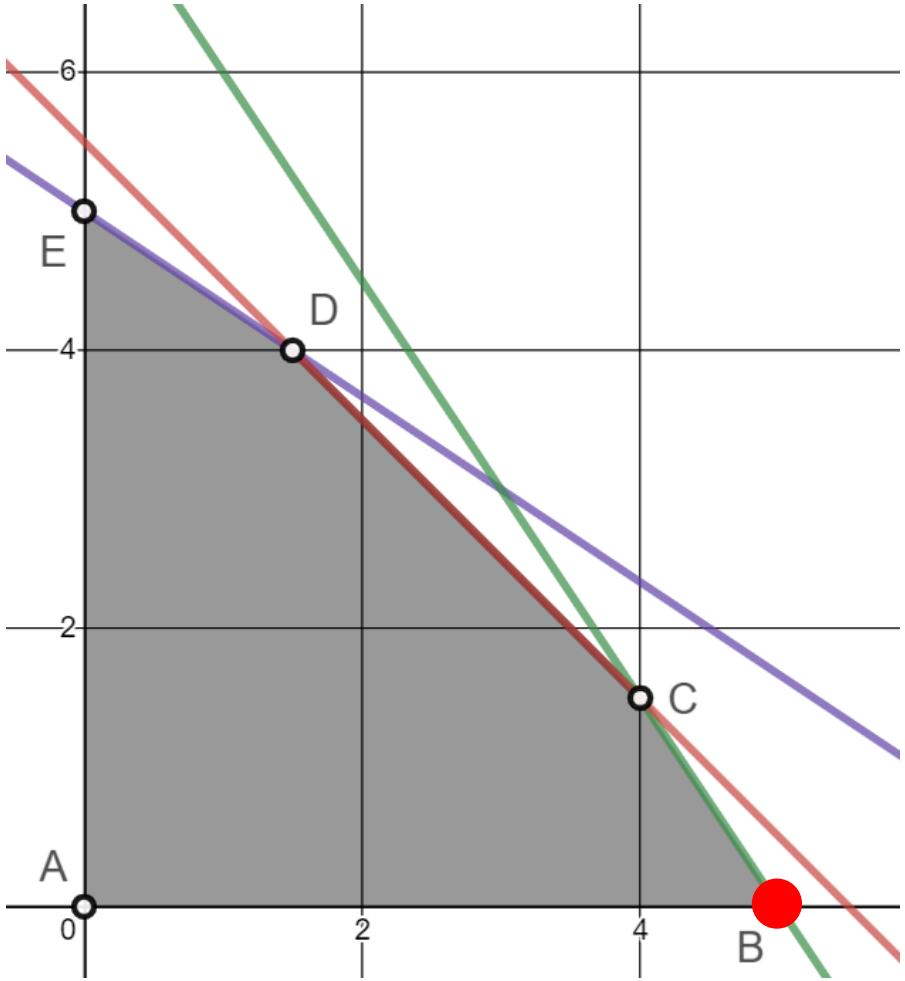
Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$

Point B $(5,0)$ has an objective value of

$$P = 5 \cdot 5 + 4 \cdot 0 = 25$$

Point E $(0,5)$ has an objective value of

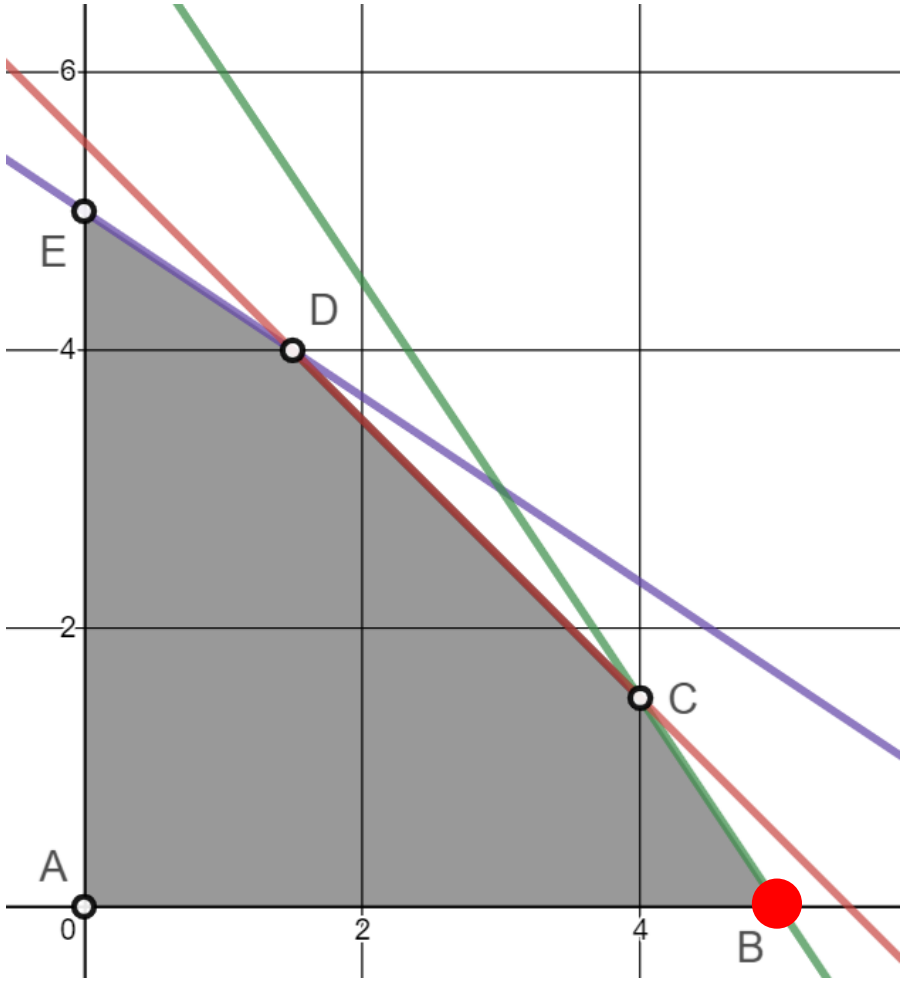
$$P = 5 \cdot 0 + 4 \cdot 5 = 20$$



$$P = 5x + 4y$$

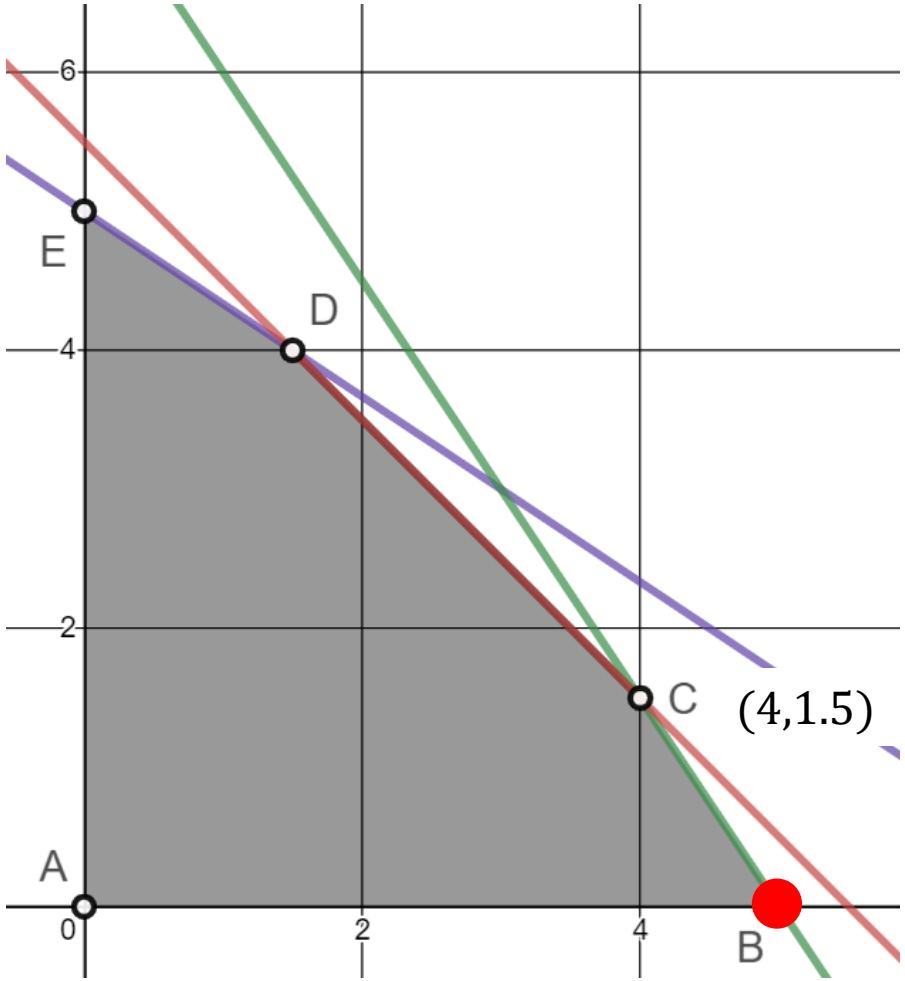
Current Point	Adjacent Points	
$A = 0$	$\boldsymbol{B = 25}$	$E = 20$

Choose B to move to!



$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$\boldsymbol{B = 25}$		

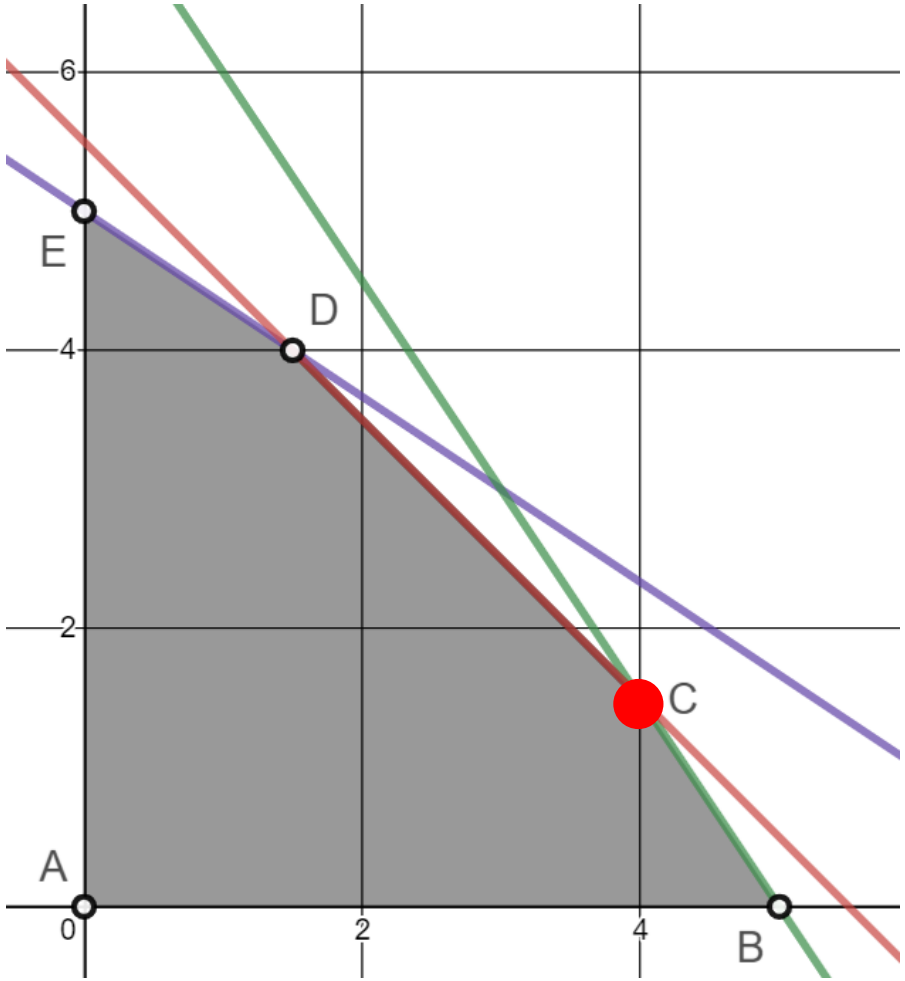


$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$

Point C $(4,1.5)$ has an objective value of

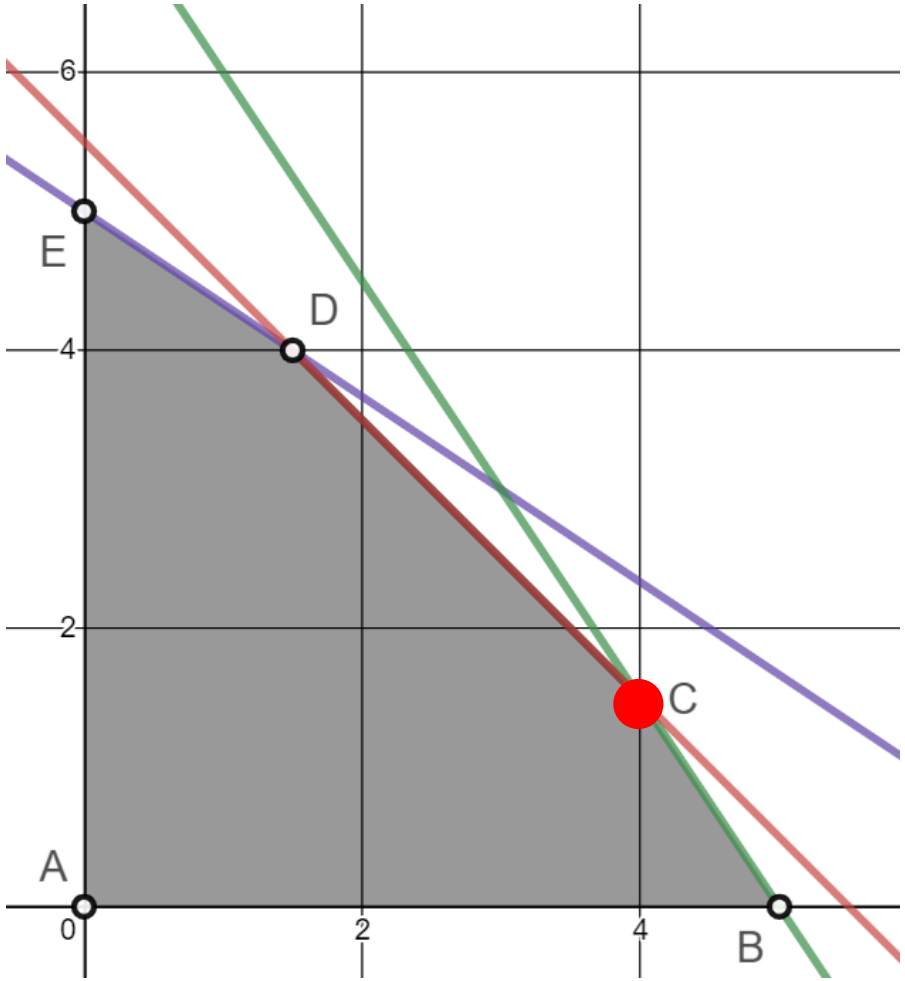
$$P = 5 \cdot 4 + 4 \cdot 1.5 = 26$$



$$P = 5x + 4y$$

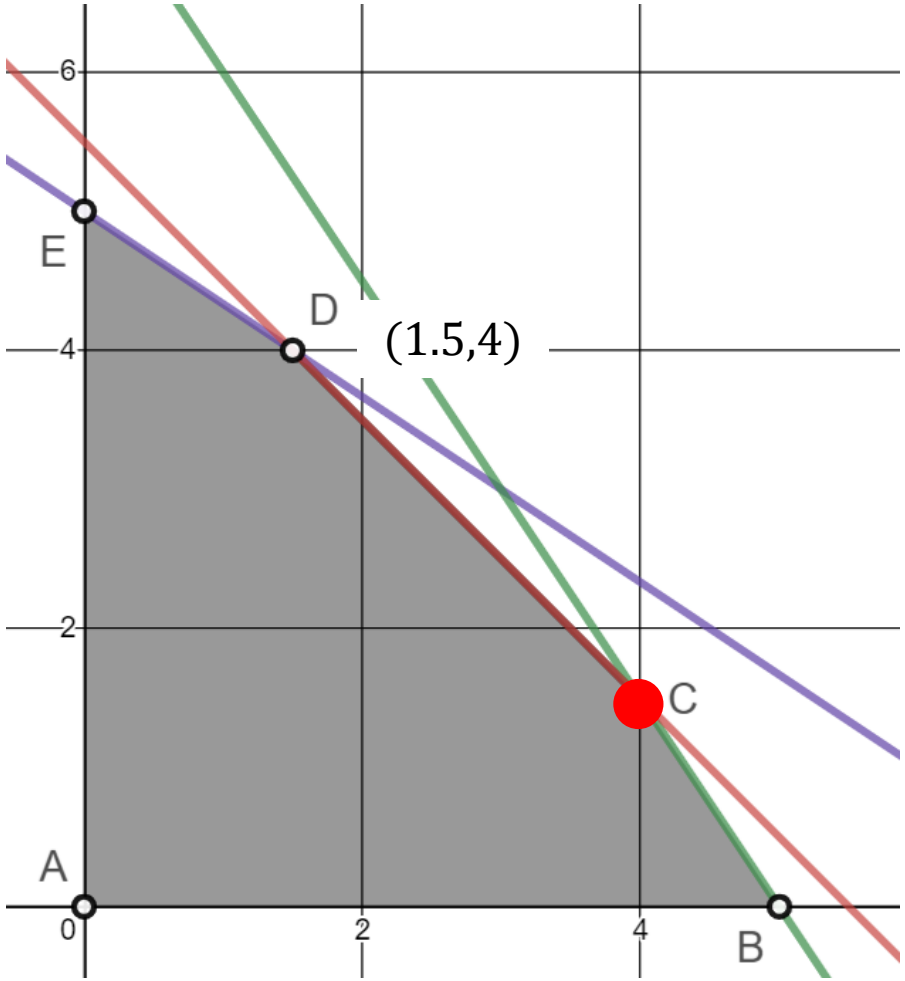
Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$\boldsymbol{C = 26}$

Choose C to move to!



$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$
$C = 26$		

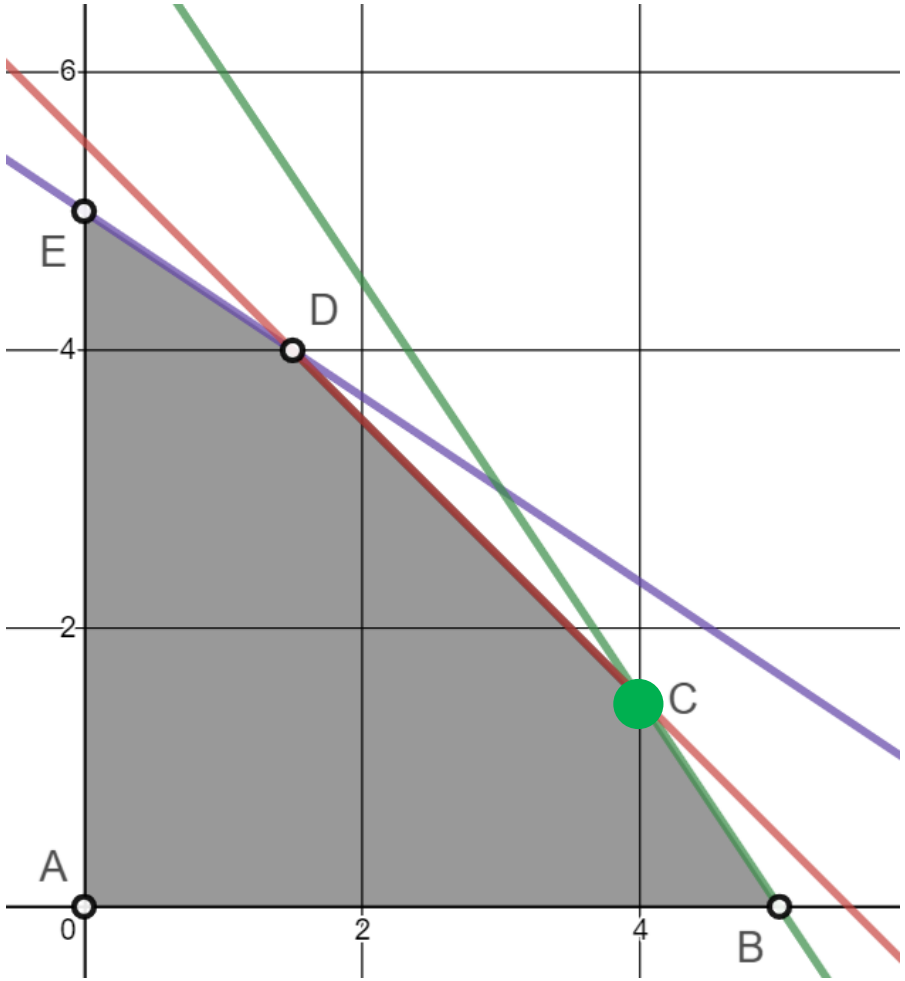


$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$
$C = 26$	$B = 25$	$D = 23.5$

Point D $(1.5,4)$ has an objective value of

$$P = 5 \cdot 1.5 + 4 \cdot 4 = 23.5$$



$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$
$\boldsymbol{C = 26}$	$B = 25$	$D = 23.5$

Cannot improve, optimal solution at C!

Higher dimensions

